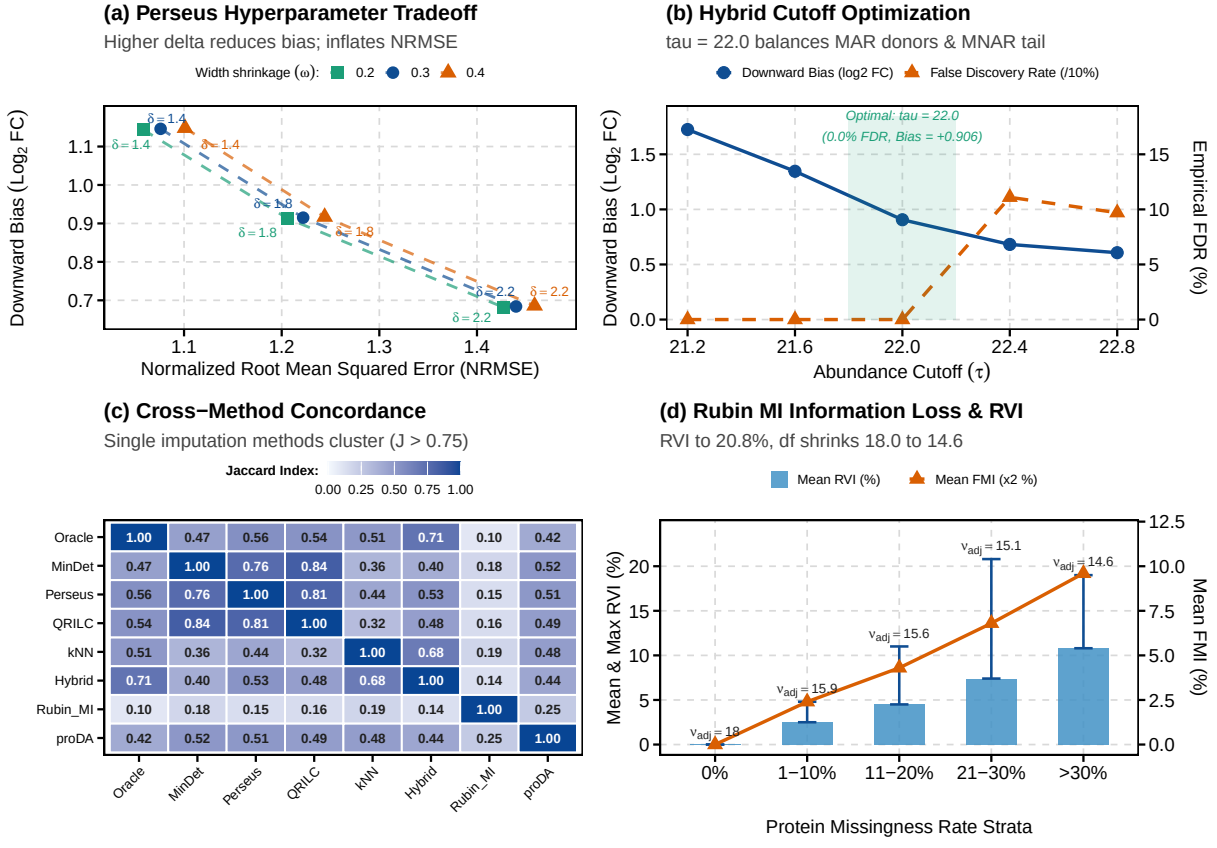
(a) Perseus Hyperparameter Tradeoff
Higher delta reduces bias; inflates NRMSE
Width shrinkage (ω): 0.2 0.3 0.4
δ = 1.4
δ = 1.8
δ = 2.2
Downward Bias (Log2 FC)
Normalized Root Mean Squared Error (NRMSE)
(b) Hybrid Cutoff Optimization
tau = 22.0 balances MAR donors & MNAR tail
Downward Bias (log2 FC)
False Discovery Rate (/10%)
Optimal: tau = 22.0
(0.0% FDR, Bias = +0.906)
Downward Bias (Log2 FC)
Empirical FDR (%)
Abundance Cutoff (τ)
(c) Cross−Method Concordance
Single imputation methods cluster (J > 0.75)
Jaccard Index:
0.00 0.25 0.50 0.75 1.00
Oracle 1.00 0.47 0.56 0.54 0.51 0.71 0.10 0.42
MinDet 0.47 1.00 0.76 0.84 0.36 0.40 0.18 0.52
Perseus 0.56 0.76 1.00 0.81 0.44 0.53 0.15 0.51
QRILC 0.54 0.84 0.81 1.00 0.32 0.48 0.16 0.49
kNN 0.51 0.36 0.44 0.32 1.00 0.68 0.19 0.48
Hybrid 0.71 0.40 0.53 0.48 0.68 1.00 0.14 0.44
Rubin_MI 0.10 0.18 0.15 0.16 0.19 0.14 1.00 0.25
proDA 0.42 0.52 0.51 0.49 0.48 0.44 0.25 1.00
(d) Rubin MI Information Loss & RVI
RVI to 20.8%, df shrinks 18.0 to 14.6
Mean RVI (%)
Mean FMI (x2 %)
Mean & Max RVI (%)
Mean FMI (%)
Protein Missingness Rate Strata
0%
1−10%
11−20%
21−30%
>30%